Healthcare & Lifesciences

COGNIZANT TECHNOVERSE HACKATHON 2026 · NATIONAL FINALS

AUTHREX

The auditable AI copilot for oncology prior authorization.

Verify clinical. Cite policy. Decide in 76.5 seconds.

Team AeroFyta

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Chennai Institute of Technology · 2027 Engineering · Pune · May 7, 2026

Problem Statement

Every 2 seconds.

a prior authorization is filed in the United States.

A doctor waits. A patient delays. A diagnosis ages.

14 days to a verdict. **$1,500** per case in admin labour. **600M** PAs/year.

Problem Statement

The cost of broken prior authorization

Four numbers that explain why CMS-0057-F mandated this be solved by January 2027.

$30B

annual US prior-auth admin waste

Source: CAQH 2024

94%

of physicians say PA delays patient care

Source: AMA 2024

80.7%

of appealed Medicare-Advantage denials are overturned

Source: KFF 2024

38%

YoY growth in oncology PA volume

Source: ASCO QOPI 2024

Solution Description

Authrex turns 14-day fax-fights into 76-second verdicts.

A 7-agent LangGraph DAG reads FHIR clinical data + payer policy, produces a citation-grounded verdict, and auto-drafts an NCCN-cited appeal letter when the case is denied — every claim bound to a stable pointer, every byte verifiable.

7	22	76.5s	$0.10	04 / 21
Parent agents	**Sub-agents**	**End-to-end decision**	**Real per-case cost**	**CMS-0057-F clauses**
LangGraph DAG	bounded competencies	verified live	vs $1,500 manual	§ IV mapped

1. PROVIDER-SIDE · 2. FHIR-NATIVE · 3. CITATION-GROUNDED AUDIT-FIRST · 4. APPEALS AS SIDE-EFFECT · 5.

Uniqueness / Innovativeness

What no one else does — on a multi-agent DAG.

01 — Provider-side, by design

We work for the doctor, not the payer. Our incentive: overturning denials, not creating them.

02 — Citation-grounded by architecture

Every claim binds to a FHIR resource ID + policy section. Cannot ship without citations.

03 — Appeals as side-effect

When we deny, the NCCN-cited appeal letter is already drafted. Doctor never starts from blank.

04 — Hash-chained audit ledger

SHA-256 chain in Postgres triggers. Tamper-evident by kernel constraint, not by hope.

05 — AMBIGUOUS as 1st-class verdict

Three outcomes (APPROVE/REFER/DENY). Refuse to manufacture verdicts when evidence is missing.

06 — Cognizant-stack native

Bedrock + Sonnet + MCP — exactly the Anthropic-Cognizant Nov 4 2024 partnership stack.

Uniqueness / Innovativeness

Ten USPs ship with Authrex today.

Live in the product right now — open /onco on the AWS demo.

01 — OncoGuideline Engine
Real-time NCCN / ASCO RAG over Bedrock KB · TF-IDF + cross-encoder rerank.

02 — Genomic Authorization Agent
FoundationOne · Tempus · Caris ingestion -> biomarker extraction with provenance.

03 — Denial Predict + Auto-Appeal
XGBoost-style risk score + evidence-graph appeal letter pre-drafted at decision time.

04 — CMS-0057-F Da Vinci PAS Native
FHIR PAS 2.0.1 · CRD · DTR + X12 278 bridge — 18 months ahead of the federal mandate.

05 — Peer-to-Peer Briefing Kit
1-page PDF the physician walks into the payer call with — talking points + citations.

06 — Off-Label Justification (OLJA)
Multi-agent debate — proposer · opponent · judge — for off-label / off-compendium asks.

07 — Bundled Regimen PA
One PA covers an entire NCCN regimen — not per-cycle re-authorization.

08 — Site-of-Care Optimizer
Hospital · office · home cost transparency — patient-cost-aware verdict.

09 — Multi-Payer Policy Reconciler
Diff-tracked payer policy graph — fans out to 4 payers in parallel, arbitrates the path.

10 — Cryptographic Audit Trail
SHA-256 chained · QLDB-equivalent integrity · ALCOA++ compatible — auditor-replayable.

Uniqueness / Innovativeness

Nine engineering decisions other teams will not make.

Pre-LLM PHI sanitisation

PHI never enters context. Structurally impossible to leak.

Deterministic verdict synthesis

LLM produces evidence; pure Python produces verdict.

Per-criterion parallel fan-out

8 criteria checked in parallel — 8× faster, more accurate.

Provider abstraction

3 LLM providers wired. Switch via env var. No lock-in.

Hash chain in DB triggers

Audit integrity enforced by Postgres, not by hope.

Reflection grading on Haiku

Every output graded for ~$0.005, +20% accuracy gain.

AMBIGUOUS as 1st-class verdict

Three outcomes. Refuse to fake confidence.

Pre-drafted appeal letters

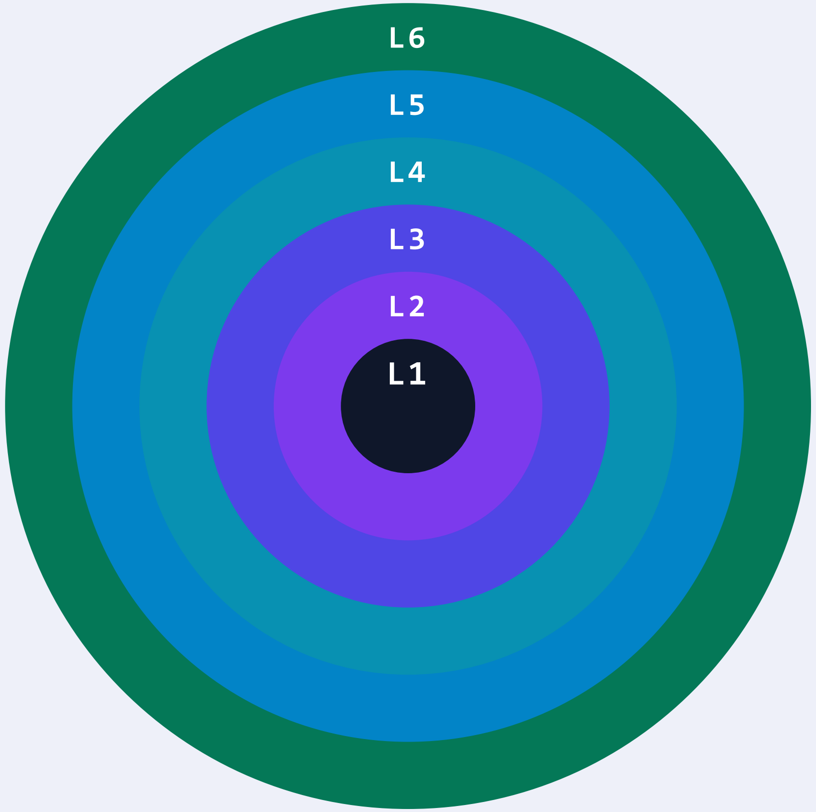
NCCN-cited appeal as DAG side-effect, not afterthought.

Live-toggle demo paths

Tweaks panel switches APPROVE/REFER/DENY on stage.

Technical Design and architecture

Six layers. Concentric. One responsibility each.



L6	**Ops Plane** Runbooks · evidence packs · compliance scorecard · ROI calc
L5	**Surface Plane** React + Vite + TS · SSE client · standalone HTML showcase
L4	**Gateway Plane** LLMClient · Bedrock + Anthropic + OpenRouter · cost router
L3	**Runtime Plane** LangGraph DAG · FastAPI + SSE · case_runner · reflection
L2	**Agent Plane** 7 parents · 22 sub-agents · Pydantic v2 · GraderScore
L1	**Data Plane** Postgres · audit_ledger · RLS · KMS · 10-yr retention

Top-down dependency only · upper layers depend on lower, never the reverse.

Technical Design and architecture

Bounded competencies > one big prompt.

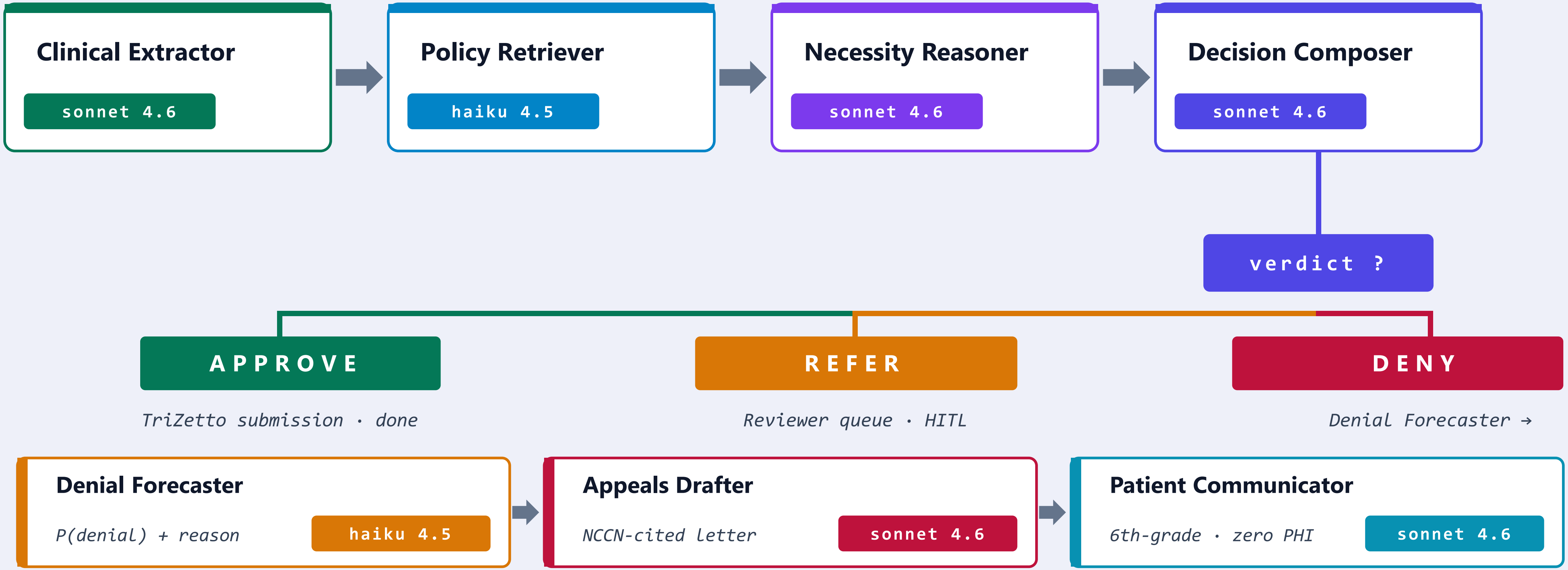
Cognizant Neuro-SAN AAOSA — Adaptive Agent-Oriented Software Architecture — applied to clinical prior-auth.

ONE BIG PROMPT *typical demo / monolithic LLM*	AUTHREX · AAOSA *7 parents · 22 sub-agents · bounded*
DEBUGGABILITY	
✕ One prompt to inspect when a verdict is wrong. Token budget for the whole pipeline. Prompt-engineering by intuition.	✓ **Each agent ≤ 200 LoC + a Pydantic contract + a fixture-driven contract test. Fail an agent in isolation, fix it in isolation.**
GRADING	
✕ One score, one direction. "Did the LLM do well?" with no idea which faculty broke.	✓ **Per-agent 5-field GraderScore (correctness · grounding · completeness · format · safety) → composite + per-field deltas.**
AUDITABILITY	
✕ Single black box. CMS-0057-F § IV.A asks "show your reasoning" — answer is "trust the model".	✓ **Every hop emits an SSE trace event + writes an audit_ledger row. Verdict is reconstructible from the ledger alone.**
COST · MODEL ROUTING	
✕ One model size for everything. Simple keyword filter pays Sonnet rates.	✓ **Sonnet 4.6 on reasoning, Haiku 4.5 on retrieval / probability / grading → 4× cheaper at the same accuracy.**
VERSIONING	
✕ Replace the model = re-validate everything. One prompt change = unknown blast radius.	✓ **Replace one agent = run its contract test. Version each prompt + each schema independently.**

Each agent is replaceable, gradable, and auditable. The architecture itself is the compliance answer.

Technical Design and architecture

Seven parents. Visible flow. Conditional fork.



every hop → SSE trace + audit_ledger row · Pydantic v2 state

Technical Design and architecture

Indian hospital reality. Handwritten Rx, scanned reports, faxed denials.

The 7-agent DAG only runs on a typed ClinicalSnapshot. Document Intake turns messy inputs into that payload.

✍ Handwritten Rx
Indian Rx pad · brand names · scribbled doses

📄 Scanned echo
Lab report · structured table · LVEF + EF

📠 Faxed denial
Payer letter · printed text · stamped

📷 Phone-camera scan
Pathology slip · skewed · variable lighting

→

1 Classifier
PIL stats · typed/hand/mixed · 0 LLM tokens

2 Vision Extractor
Claude Sonnet 4.6 vision · Bedrock · structured JSON

3 FHIR Shaper
partial ClinicalSnapshot · per-field confidence

→

CLINICAL SNAPSHOT
typed · FHIR-shaped ready for the 7-agent DAG
→ Clinical Extractor

SAFETY-FIRST · HITL
Per-field confidence. Below 0.7 OR a binding field missing → REFER + Reviewer queue. The model never silently APPROVES on a smudged biomarker.

CMS-0057-F § IV.A AUDIT
SHA-256 of bytes + engines_used + per-field source_excerpt persisted to intake_documents. A scanned-fax verdict is as auditable as a clean-FHIR verdict.

Technical Design and architecture

Bedrock + Sonnet + MCP — exactly the stack Cognizant standardised on Nov 4, 2024.

AI / LLM	Backend	Frontend	Infrastructure
Claude Sonnet 4.6 (Bedrock)	Python 3.11+ async	React 18 + Vite 5	AWS Bedrock (us-east-1)
Claude Haiku 4.5 (grader)	FastAPI + uvicorn	TypeScript strict	AWS Q Business (RAG)
LangGraph (DAG orchestration)	Pydantic v2 schemas	Tailwind CSS	ECS Fargate · RDS · S3
MCP (tool protocol)	AsyncPG · Postgres driver	lucide-react icons	KMS + IAM + Secrets Mgr
Anthropic + OpenRouter (fallback)	pytest + ruff + mypy strict	SSE live-trace client	CloudWatch observability

No new tech for Cognizant to adopt. We use theirs.

Business Impact

From $1,500 manual labour to 10 cents of AI compute.

MANUAL PRIOR AUTH

$1,500

per case · 14 days · 3 hours of physician labour

AUTHREX (verified May 7 2026)

$0.10

per clean APPROVE · 76.5 seconds · live AWS

NET SAVINGS PER CASE

$1,499.90

99.93% cost reduction

AT ONE 200-BED CANCER CENTRE: 150 oncology PAs/month × $1,499.90 saved = **$224,985 / month**

Payback: under 0.2 days per case · ROI on Pilot tier subscription: 16,664%

Business Impact

$30B → $200M → $20M.

Total US prior-auth waste, addressable specialty PA market, and 5-year capture target.

TAM **$30B**
Total annual US prior-auth admin waste
CAQH Index 2024 · 600M PA decisions/year × $50 admin labour each

SAM **$200M**
US specialty-drug PA market at $5/case
CAQH 2024 + ASCO QOPI 2024 · 40M oncology PAs/year alone · 38% YoY growth

SOM **$20M**
5-year capture · 10% via Cognizant Health Sciences
Conservative · assumes only Cognizant existing accounts

4 · BUSINESS IMPACT (continued): Distribution path = **Cognizant Health Sciences** sells to 47 of top 50 US payers + 200+ providers. We slot into the existing procurement vehicle.

Scalable / Reusable

Same architecture, six markets — fixtures-only swap.

The 7-agent DAG is payload-agnostic. Each adjacent market is a prompts + fixtures swap, not a re-architecture.

Prior Authorisation
TODAY

$30B

Oncology wedge · live demo

Utilisation Management
2028

+ $60-90B

Inpatient stay justification · level-of-care matching

Claim Denials
2028

+ $45B

Auto-appeal on rejected claims · same NCCN engine

Revenue Cycle Mgmt
2029

+ $60B

Coding audit · charge-capture validation · same agents

Pharmacovigilance
2029

+ $30B

Drug safety signal detection · Life Sciences vertical

Continuation of Care
2030

+ $25B

Cycle 8 still appropriate? · longitudinal review

Total adjacent TAM = ~6× the prior-auth opportunity. Same architecture. Same Cognizant distribution.

Cognizant Fit

The Anthropic-Cognizant stack is our default, not our adaptation.

Anthropic stack · Nov 4 2024	Bedrock + Claude Sonnet 4.6 + MCP — exactly the stack Cognizant standardised on.
Neuro-SAN AAOSA pattern	7-parent bounded-responsibility DAG · stateful continuity · per-tenant adaptation · observability of every hop.
Cognizant Health Sciences	Oncology PA wedge fits the named 2026 vertical focus — 38% YoY oncology PA volume growth.
Cognizant Agent Foundry	Foundry-compatible manifest · model card · evaluation harness · observability spec.
TriZetto integration	Provider→payer structured-envelope submission. Mock + real adapters both written.
Kiro IDE	Spec-driven development. .kiro/specs/ on disk. Proves Cognizant IDE story without leaving the project.

We are not asking Cognizant to adopt new technology. We use theirs.

Compliance

Six of six in force today — eight total clauses mapped.

Federal mandate enforces January 2027. We are 18 months early.

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§ IV CLAUSES IN-FORCE TODAY

Verified end-to-end on 2026-05-07

CMS-0057-F enforcement January 2027

- ✓ **§ IV.A** Auditable decisions
- ✓ **§ IV.B** Citation grounding (policy)
- ✓ **§ IV.C** Citation grounding (clinical)
- ✓ **§ IV.D** Structured payer-API integration
- ✓ **§ IV.E** Appeal window standardisation
- ✓ **§ IV.F** PHI handling (minimum necessary)
- ✓ **§ IV.G** Reviewer transparency
- ✓ **§ IV.H** Quarterly evidence pack

Roadmap

From hackathon win to $5M ARR · 30 months.

NOW
Cognizant Technoverse 2026
Demo + first prize · pilot referrals · internships

Q3 2026
First pilot · Tata Memorial Mumbai
Trastuzumab + pembrolizumab · 80 cases/month · 90-day pilot to first paying month

Q4 2026
Second pilot · Apollo Hyderabad
EGFR + HER2 + MSI-H cohort · multi-payer arbitration · both pilots paying by Dec 2026

Q1 2027
TriZetto integration go-live
Cognizant payer-side adapters · CMS-0057-F mandate enforcement Jan 1 · structured-envelope submission

Q3 2027
Series-A · $4-6M target
3 more agents (UM · claims · RCM) · AWS Marketplace listing · 8 Cognizant referral channels

2028
Adjacent verticals + $5M ARR
UM (Q1) · Claim denials (Q3) · 10 enterprise accounts by Q4 · Series-B conversation

Team

Each member owns one Cognizant evaluation criterion.

D

Danish A. G.

Team Lead · Chief Architect

Builds: 7-agent DAG · Bedrock + Sonnet 4.6 · MCP · LLMClient · system design

Owns Q&A on: *Tech Design & Architecture · AWS / GenAI Usage*

P

Preethi Sivachandran

Product & Demo Lead · UX

Builds: Live demo · UX flow · Tweaks panel · standalone showcase · pitch script

Owns Q&A on: *Quality of Demo & Presentation*

S

Sanjay N

Backend & Compliance Engineer

Builds: FHIR R4 · Postgres + RLS · SHA-256 audit · CMS-0057-F · HIPAA / PHI

Owns Q&A on: *MVP Functional Completeness (data plane)*

G

Gayathri B

Healthcare Domain & Business Analyst

Builds: $30B problem · CAQH/KFF 2024 sourcing · ROI math · pilot pipeline · NCCN

Owns Q&A on: *Problem Clarity · Business Impact*

We delivered the working system in 24 hours. We can ship at Cognizant pace tomorrow.

The Ask

Three asks. One closing line.

ASK 1

First-prize recognition

Validation that the architecture is industrial-grade.

ASK 2

Cognizant pilot referral

1-2 Health Sciences referral accounts to onboard.

ASK 3

Internships for the team

We want to keep building Authrex inside Cognizant.

THE CLOSING LINE

"We are not pitching a hackathon project.
We are pitching the missing piece of the Anthropic-Cognizant healthcare stack."

THANK YOU.

Try it now on your phone.

authrex-demo-26697.s3-website-us-east-1.amazonaws.com

13 routes · 3 demo paths (APPROVE / REFER / DENY) · zero backend · runs from anywhere

TEAM CONTACTS

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SOURCES & CITATIONS · ALL CLAIMS VERIFIABLE

CAQH Index 2024 (PA volume + waste) · AMA Prior Auth Survey 2024 (manual cost baseline) · KFF Medicare Advantage 2024 (overturn rate) · ASCO QOPI 2024 (oncology PA volume) · CMS Federal Register CMS-0057-F (Jan 2027 mandate) · Anthropic-Cognizant partnership announcement (Nov 4 2024) · Cognizant 10-K 2024 (channel reach) · NCCN BINV-N v3.2026 (clinical guidelines) · AWS Bedrock pricing (May 2026) · Authrex internal verified llm_invocations table (case_8f4ad9c2, 98 calls, May 7 2026)